Indus Insights – Aptitude Test

The test consists of 3 sections with sectional cut-offs:

1. Problem Solving
2. Data interpretation
3. Critical Reasoning

Sample Questions

Problem Solving (*Suggested time per question: 5 minutes*)

1. A company wants to sell its products to potential customers over the phone. To do this, it needs to purchase a list of potential customers, with their phone numbers. Their list vendor has given the company two options – List X and List Y. List Y is more expensive but is supposedly of "better quality". List X costs Cx per phone number. List Y costs Cy per phone number. The company tested the two lists and found that List X had a "call fail rate" (i.e. % of attempted calls that didn't get connected) of Fx. For List Y, this was Fy. Of the calls that did get connected, customers sourced through List X purchased with a propensity Px (i.e. % of connected calls that led to a purchase). For List Y, this was Py. When customers did purchase something, List X customers generated sales of Sx per purchase and List Y customers generated sales of Sy per purchase. It can be assumed that the margin on all sales is the same. Ignoring call center costs (phone calls, agent time, etc), **the company should opt for List Y if**:

   (Assume that both lists have enough phone numbers for the company)

   a. $\frac{Cx}{Cy} < \frac{Fx \times Px \times Sx}{Fy \times Py \times Sy}$

   b. $\frac{Cx}{Cy} > \frac{Fx \times Px \times Sx}{Fy \times Py \times Sy}$

   c. $\frac{Cx}{Cy} > \frac{(1-Fx) \times Px \times Sx}{(1-Fy) \times Py \times Sy}$

   d. $\frac{Cx}{Cy} < \frac{(1-Fx) \times Px \times Sx}{(1-Fy) \times Py \times Sy}$

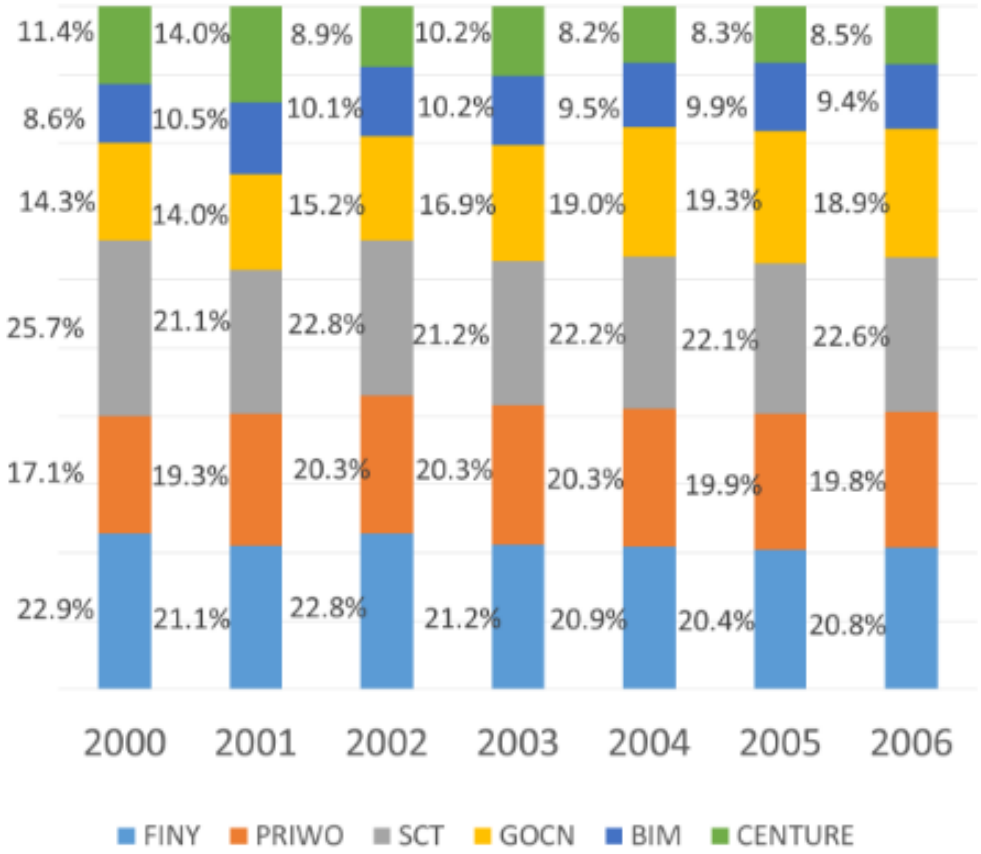
2. An oil drilling company is evaluating the decision of improving an oil rig or not. The rig in its current condition is operational. It generates 400 barrels of oil per functional day and is in-repair (i.e. isn't functional) 35 days in a year. The company can make an investment of $20 Million (i.e. $20,000,000) to improve the rig. If this is done, it would increase the production to 600 barrels of oil per functional day and the rig would be in-repair 15 days in a year. The company wants to recoup its investment in 5 years (assume each year has 365 days). For every barrel that the oil rig produces, the company makes a profit of P. **What is the minimum value of P above which the company should make the investment?**

   (Ignore any others costs such as people idle time, costs in packing/cleaning oil, running costs of the rig, etc)

   a. $ 255
   b. $ 51
   c. $ 69
   d. $ 345

Data Interpretation (*Suggested time per question: 1.5 minutes*)

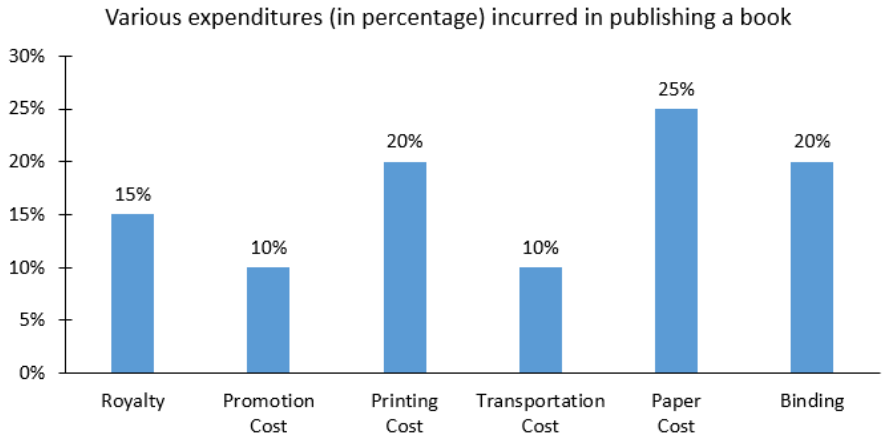
3. The following chart gives the revenue share of different companies in the tech sector in Motherland:



In 2001, the company that grew quickest grew by 100%, what was the growth rate of company that had the least growth rate?

a. 50%
b. 25%
c. 20%
d. 33%

4. The following bar chart shows the percentage distribution of the expenditure incurred in publishing a book:



The price of the book is marked 20% above the cost price. If the marked price of the book is INR 180, then what is the cost of the paper used in a single copy of the book?

a. INR 36
b. INR 37.5
c. INR 42
d. INR 44.25

Critical Reasoning (*Suggested time per question: 2 minutes*)

5. In a certain state, hunting permits require the hunter to be at least 18 years old, possess a valid driver's license or state identification, and have completed a safety program within the past 5 years. The hunter must also sign a legal document pledging not to consume alcohol while hunting.

   Which of the following can be correctly inferred from the above statements?

   a. All states have the same requirements for hunting permits
   b. Hunters under the age of 21 don't need to sign the alcohol pledge because they are not legally allowed to drink alcohol under any circumstances
   c. If a person last completed the safety program six years ago, he will have to complete the program again before he can be eligible for a permit
   d. Someone who isn't 18 years old cannot obtain a hunting permit in this state
   e. Hunting is such a dangerous activity that state controls and requirements are necessary to ensure that nobody gets hurt

6. The number of children who are obese – that is, who have more body fat than do 85 percent of children their age – is steadily increasing, according to four major studies conducted over the past 15 years.

   If the finding reported above is correct, it best supports which of the following assertions?

   a. When four major studies all produce similar results, those studies must be accurate
   b. Children have been progressively less physically active over the past 15 years
   c. The number of children who are not obese increased over the past 15 years
   d. Over the past 15 years, the number of children who are underweight has declined
   e. The incidence of obesity in children tends to increase as the children grow older

Answer Key

1. C
2. B
3. D
4. B
5. C
6. C